

Update Joint Modeling 07/10

Overview methods

Table 1: Configuration of the methods.

Method	Configuration
Classical ML	<code>lme4::lmer</code> , ML (REML = FALSE); Wald z test on β_3
MI + closed form	<code>mice 21.pmm</code> , 3 imputations, <code>maxit</code> = 10; closed form (CbC) per imputation, pooled with Rubin’s rules; Wald z test on β_3
Reweighting	Closed form, reweighted until $\max \Delta\beta < 10^{-6}$, at most 30 iterations; Wald z test on β_3
LSPIM	Pseudo-score GEE model, Holm-adjusted interaction test; only run for $n \leq 50$

Table 2: Definition of “not converged” per method.

Method	“Not converged” when
Classical ML	optimizer code $\neq 0$ or an <code>lme4</code> convergence message (e.g. $\max \text{grad} > 0.002$)
MI + closed form	never; a failed fit in any imputation counts as an error
Reweighting	the loop reaches 30 iterations without meeting the $\Delta\beta$ criterion
LSPIM	any of its GEEs does not converge

Each replicate gets exactly one of five convergence statuses, checked in this order:

Table 3: Convergence statuses.

Status	Meaning
Error	The fit failed (no estimates); excluded from all metrics.
Not converged	Fit returned estimates, but the method’s convergence criterion (table above) was not met.
Converged (singular)	Converged, but the random-effects covariance is at the boundary (smallest eigenvalue $\leq 10^{-6}$, or D repaired to positive definite).
Converged (warning)	Converged and not singular, but a warning was raised (e.g. <code>lme4</code> ’s boundary message at its looser tolerance; for reweighting a D repair or NaNs in an intermediate iteration).
Converged	Converged, not singular, no warning.

- All tests at $\alpha = 0.05$; coverage of the 95% Wald interval.
- All statuses except “error” are kept in all metrics.

 Missing results

All replicates errored for: LSPIM (48 scenarios). These rows are left out of all tables and plots.
LSPIM: package **geessbin** was not installed on the computer that ran the simulation; a rerun is pending.

Simulation setup

Linear model, $t_j = j = 0, \dots, 11$:

$$y_{ij} = \beta_0 + \beta_1 \text{Treatment}_i + \beta_2 t_j + \beta_3 \text{Treatment}_i \times t_j + b_{0i} + b_{1i} t_j + \varepsilon_{ij}$$

Log model, $t_j = j = 0, \dots, 11$:

$$y_{ij} = \beta_0 + \beta_1 \text{Treatment}_i + \beta_2 \log(1 + t_j) + \beta_3 \text{Treatment}_i \times \log(1 + t_j) + b_{0i} + b_{1i} \log(1 + t_j) + \varepsilon_{ij}$$

In both models $(b_{0i}, b_{1i})^\top \sim N(0, D)$ with $D = \begin{pmatrix} d_{11} & d_{12} \\ d_{12} & d_{22} \end{pmatrix}$ and $\varepsilon_{ij} \sim N(0, \sigma^2)$.

Table 4: Data-generating parameters; a / b / c gives the values of the large, small and null effect.

Effect	Symbol	Linear	Log
Intercept	β_0	2.4562	2.4562
Treatment effect	β_1	0	−2.42 / −0.385 / 0
Time effect	β_2	0.2792	1.2359
Treatment $\times$ time	β_3	0.22 / 0.035 / 0	1.948 / 0.310 / 0
Random intercept variance	d_{11}	7.3174	7.3174
Random slope variance	d_{22}	0.2239	4.3875
Random effects covariance	d_{12}	−0.4985	−2.2067
Residual variance	σ^2	3.1508	3.1508

- Linear parameters taken from the toenail data ($\beta_3 = 0.035$); $\beta_3 = 0.22$ added as a larger effect.
- Log parameters: β_2 , d_{22} and d_{12} rescaled by $11/\log 12$, so the rise and the slope variance at $t = 11$ match the linear scenario.
- Log crossing: arms cross at $t \approx 2.5$; the treatment difference at $t = 11$ equals that of the linear scenario with $\beta_3 = 0.22$ (large) or 0.035 (small).
- Log null: identical arms ($\beta_1 = \beta_3 = 0$).

Table 5: Simulation design.

Variable	Value
Number of clusters	10, 20, 50, 100, 1000
Number of repeated measures	12
Missingness pattern	“Half-missing” (HM): half of the clusters have no missingness, the other half have a uniform probability to keep 1–11 observations; dropout is monotone. “Three observations minimum” (3O): as half-missing, but with at least three observations per cluster.
Simulations per scenario	5000

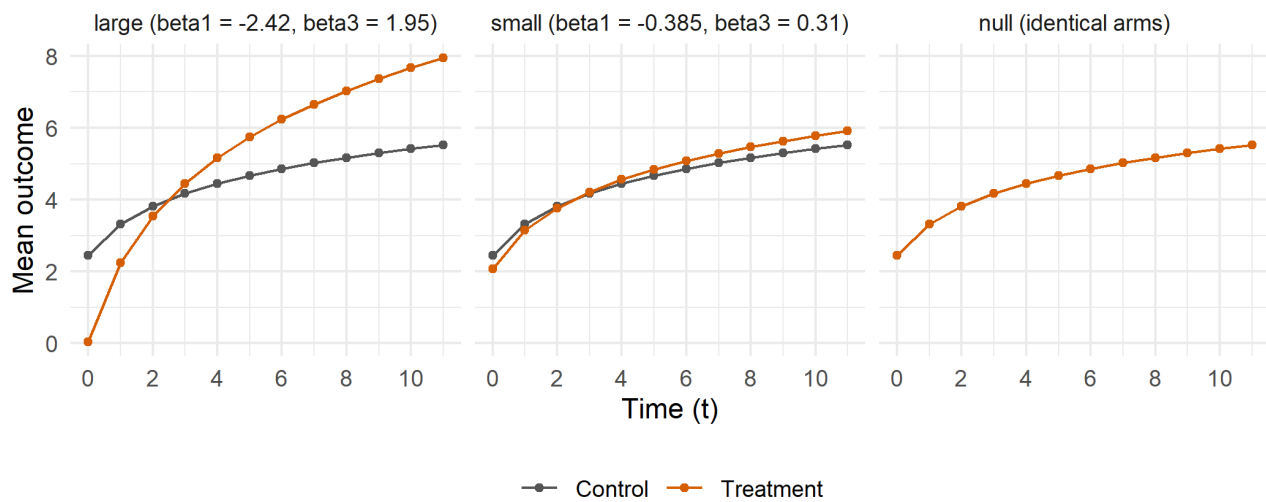


Figure 1: Mean profiles per arm for the log time trend, from the simulation parameters.

Results - linear time trend

Mean squared error of β_3

Table 6: MSE of beta3 (HM = half-missing, 3O = three observations minimum; columns = true beta3).

n	Method	HM: 0.22	HM: 0.035	HM: 0	3O: 0.22	3O: 0.035	3O: 0
10	Classical ML	0.137	0.138	0.135	0.125	0.125	0.126
10	MI + closed form	0.122	0.119	0.117	0.113	0.110	0.111
10	Reweighting	0.136	0.138	0.134	0.124	0.125	0.125
20	Classical ML	0.0640	0.0636	0.0673	0.0595	0.0631	0.0627
20	MI + closed form	0.0627	0.0621	0.0651	0.0588	0.0606	0.0608
20	Reweighting	0.0650	0.0643	0.0684	0.0594	0.0630	0.0626
50	Classical ML	0.0245	0.0253	0.0248	0.0241	0.0232	0.0236
50	MI + closed form	0.0255	0.0266	0.0258	0.0254	0.0240	0.0245
50	Reweighting	0.0249	0.0256	0.0252	0.0241	0.0232	0.0236
100	Classical ML	0.0123	0.0128	0.0122	0.0119	0.0117	0.0115
100	MI + closed form	0.0130	0.0134	0.0131	0.0126	0.0123	0.0121
100	Reweighting	0.0125	0.0129	0.0124	0.0119	0.0118	0.0115
1000	Classical ML	0.00129	0.00125	0.00125	0.00117	0.00119	0.00121
1000	MI + closed form	0.00141	0.00135	0.00136	0.00126	0.00131	0.00131
1000	Reweighting	0.00132	0.00127	0.00127	0.00117	0.00119	0.00121

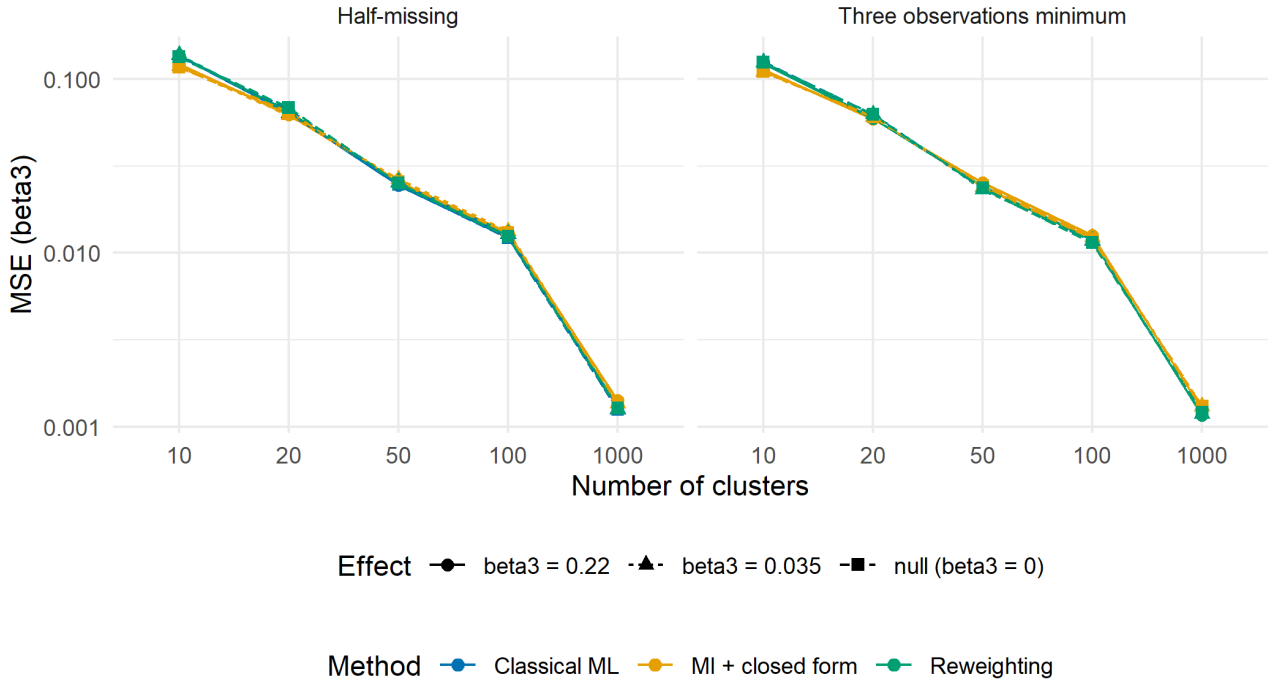


Figure 2: MSE of beta3 for the linear time trend (log scale).

- MSE falls with the number of clusters at the same rate for all methods; the effect size and dropout mechanism hardly matter.
- MI has the lowest MSE at small n ; from $n = 50$ onwards classical ML and reweighting are slightly better and similar to each other.

Type I error

Table 7: Type I error rate of the interaction test ($\beta_3 = 0$).

n	Method	HM: 0	3O: 0
10	Classical ML	0.154	0.144
10	MI + closed form	0.112	0.0924
10	Reweighting	0.102	0.101
20	Classical ML	0.0966	0.0958
20	MI + closed form	0.0802	0.0764
20	Reweighting	0.0826	0.0792
50	Classical ML	0.0616	0.0616
50	MI + closed form	0.0566	0.0552
50	Reweighting	0.0582	0.0564
100	Classical ML	0.0560	0.0556
100	MI + closed form	0.0528	0.0488
100	Reweighting	0.0532	0.0530
1000	Classical ML	0.0480	0.0500
1000	MI + closed form	0.0536	0.0504
1000	Reweighting	0.0480	0.0488

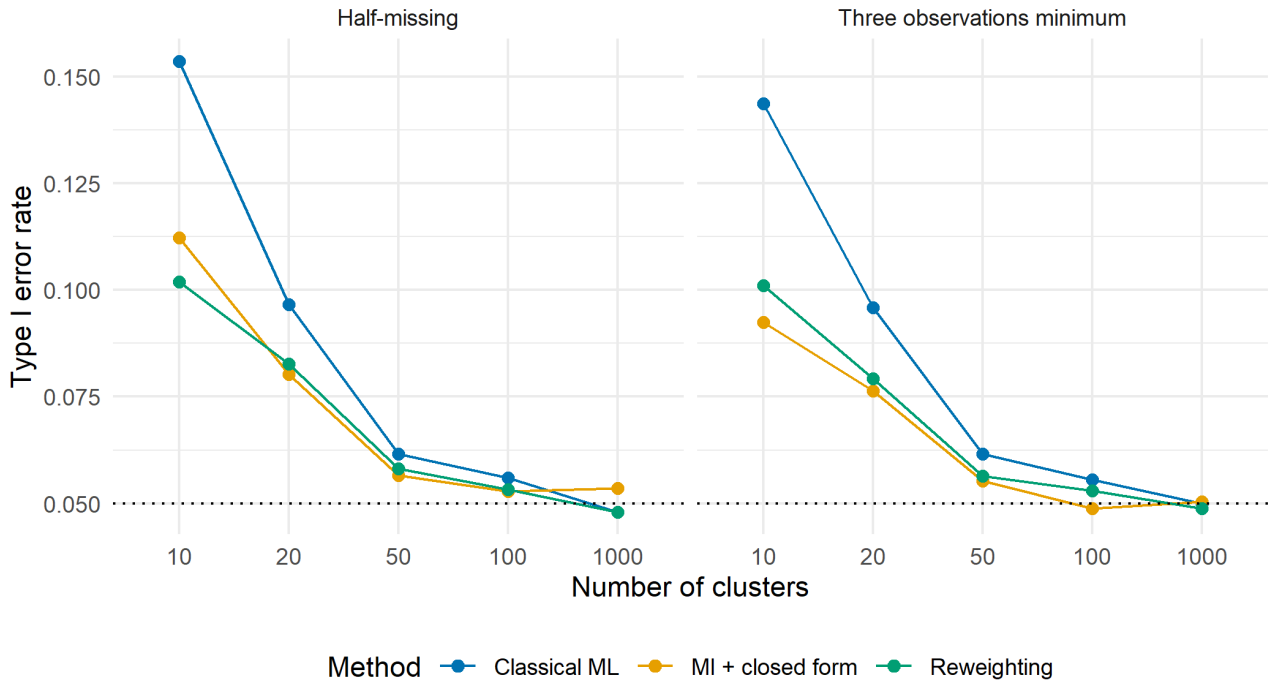


Figure 3: Type I error rate for the linear time trend (dotted line: nominal 0.05).

- All methods are anti-conservative at small n ; classical ML is inflated the most, MI and reweighting are similar.
- From $n = 100$ onwards all methods are close to nominal.

Power

Table 8: Power of the interaction test.

n	Method	HM: 0.22	HM: 0.035	3O: 0.22	3O: 0.035
10	Classical ML	0.212	0.154	0.198	0.149
10	MI + closed form	0.151	0.106	0.131	0.0970
10	Reweighting	0.158	0.103	0.145	0.104
20	Classical ML	0.196	0.0908	0.195	0.0994
20	MI + closed form	0.166	0.0746	0.153	0.0710
20	Reweighting	0.171	0.0712	0.168	0.0842
50	Classical ML	0.306	0.0714	0.314	0.0626
50	MI + closed form	0.269	0.0724	0.263	0.0584
50	Reweighting	0.285	0.0650	0.297	0.0588
100	Classical ML	0.532	0.0682	0.537	0.0680
100	MI + closed form	0.469	0.0674	0.468	0.0588
100	Reweighting	0.512	0.0640	0.529	0.0656
1000	Classical ML	1.00	0.162	1.00	0.175
1000	MI + closed form	1.00	0.151	1.00	0.159
1000	Reweighting	1.00	0.158	1.00	0.173

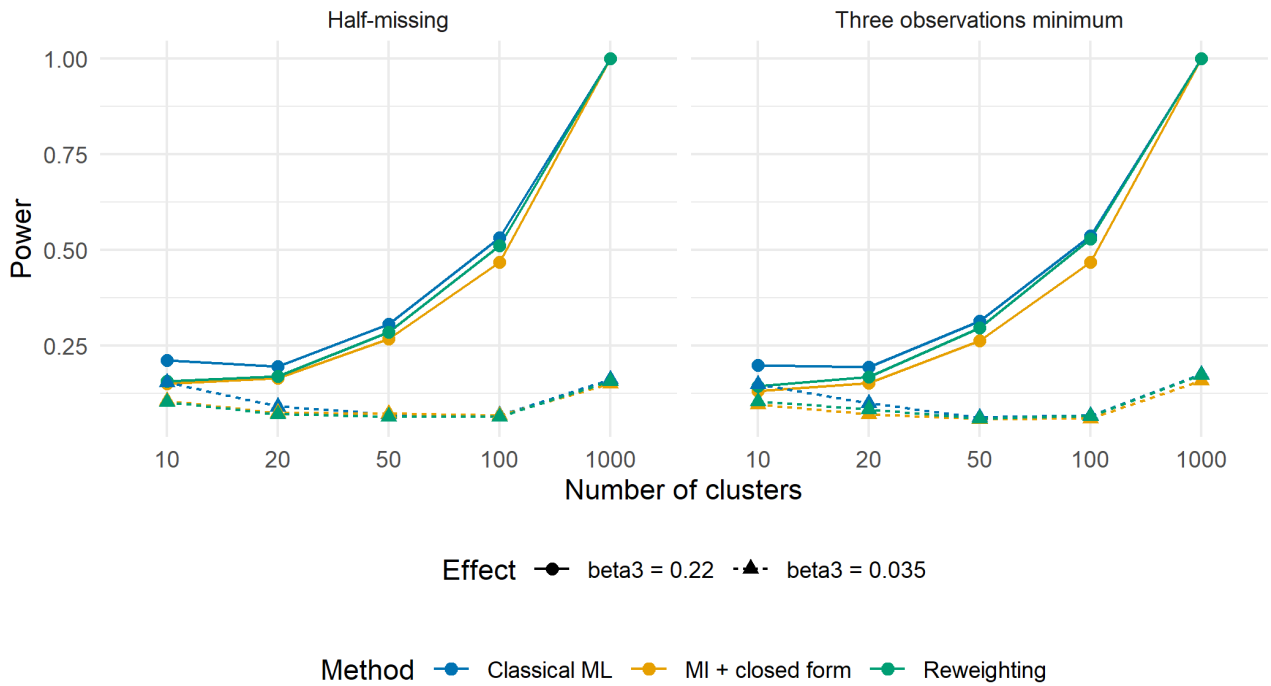


Figure 4: Power for the linear time trend.

- Classical ML has the highest power, then reweighting, then MI; at small n this ordering partly mirrors the type I error inflation.
- For $\beta_3 = 0.035$ power stays low even at $n = 1000$; for $\beta_3 = 0.22$ all methods reach full power at $n = 1000$.

Coverage of β_3

Table 9: Coverage of the 95% Wald interval for beta3.

n	Method	HM: 0.22	HM: 0.035	HM: 0	3O: 0.22	3O: 0.035	3O: 0
10	Classical ML	0.843	0.844	0.846	0.858	0.850	0.856
10	MI + closed form	0.894	0.893	0.888	0.908	0.905	0.908
10	Reweighting	0.897	0.897	0.898	0.907	0.898	0.899
20	Classical ML	0.907	0.911	0.903	0.920	0.907	0.904
20	MI + closed form	0.921	0.926	0.920	0.932	0.928	0.924
20	Reweighting	0.921	0.929	0.917	0.930	0.919	0.921
50	Classical ML	0.942	0.934	0.938	0.937	0.943	0.938
50	MI + closed form	0.942	0.934	0.943	0.942	0.947	0.945
50	Reweighting	0.943	0.939	0.942	0.937	0.947	0.944
100	Classical ML	0.944	0.943	0.944	0.943	0.941	0.944
100	MI + closed form	0.946	0.940	0.947	0.948	0.950	0.951
100	Reweighting	0.948	0.946	0.947	0.944	0.945	0.947
1000	Classical ML	0.950	0.952	0.952	0.955	0.947	0.950
1000	MI + closed form	0.943	0.950	0.946	0.956	0.949	0.950
1000	Reweighting	0.950	0.953	0.952	0.956	0.946	0.951

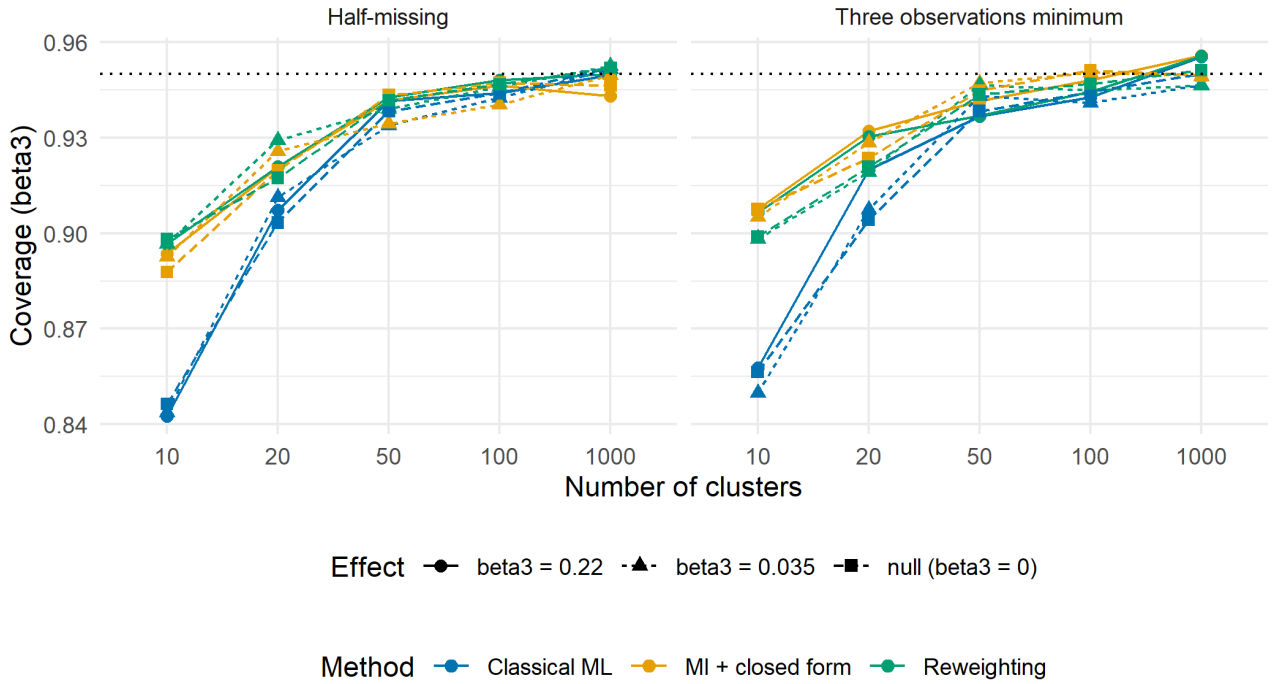


Figure 5: Coverage of the 95% interval for beta3 for the linear time trend (dotted line: 0.95).

- Undercoverage at small n for all methods; classical ML is worst, MI and reweighting are similar.
- Coverage is close to nominal from $n = 50$ onwards.

Convergence

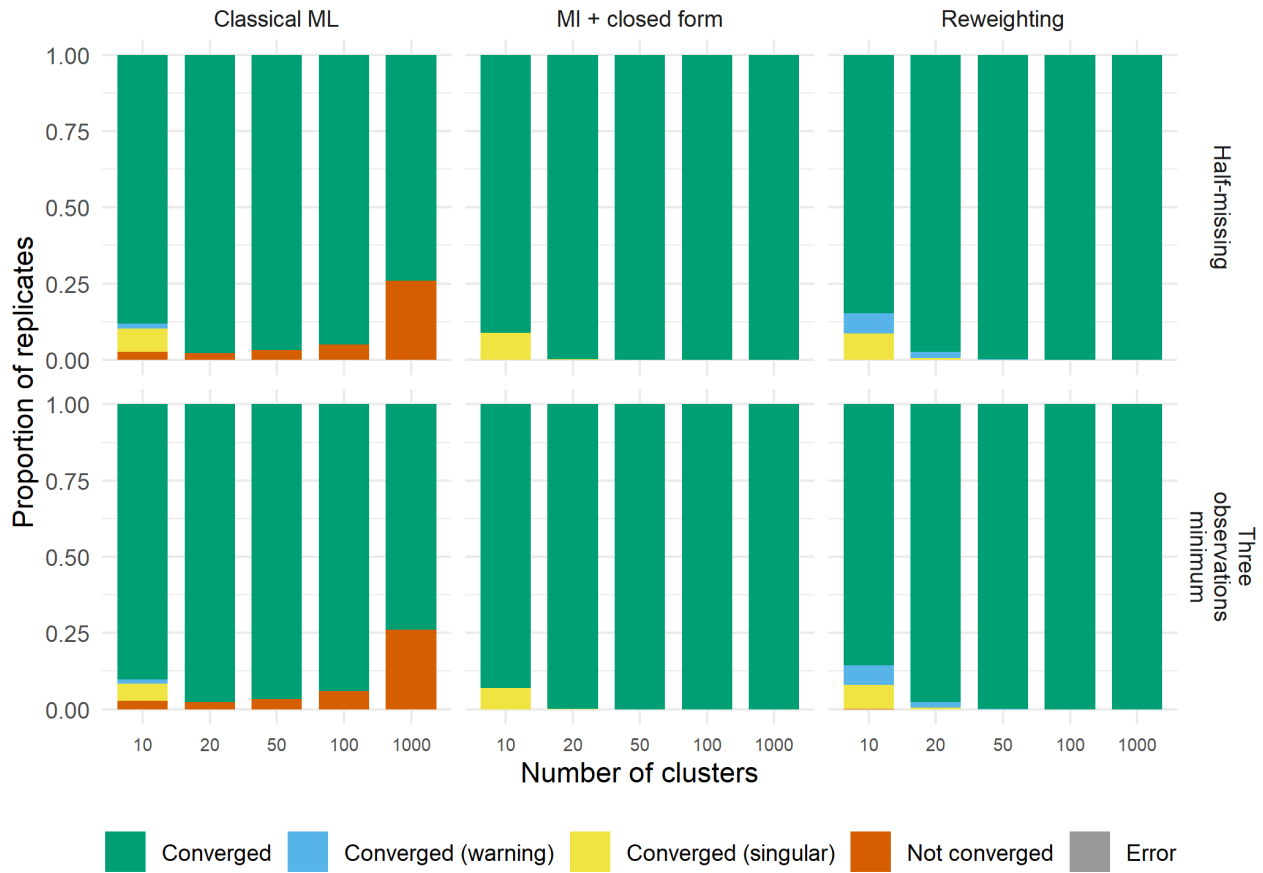


Figure 6: Convergence status for the linear time trend, pooled over effect sizes.

- Singular fits occur at $n = 10$ for all methods at a similar rate and almost vanish from $n = 20$.
- Reweighting and MI essentially always converge; classical ML has a small “not converged” share at every n , which grows sharply at $n = 1000$.
- This is a false positive of lme4’s absolute gradient check, which scales with the sample size: the flagged fits give the same estimates, coverage and power as the other fits and are kept in all metrics.

Computation time

Table 10: Median computation time per replicate (seconds).

n	Method	HM: 0.22	HM: 0.035	HM: 0	3O: 0.22	3O: 0.035	3O: 0
10	Classical ML	0.0180	0.0190	0.0200	0.0200	0.0190	0.0190
10	MI + closed form	1.15	1.15	1.15	1.16	1.16	1.16
10	Reweighting	0.0560	0.0560	0.0560	0.0620	0.0620	0.0620
20	Classical ML	0.0190	0.0190	0.0190	0.0190	0.0200	0.0200
20	MI + closed form	1.26	1.26	1.26	1.27	1.27	1.27
20	Reweighting	0.110	0.111	0.111	0.123	0.125	0.124
50	Classical ML	0.0240	0.0240	0.0230	0.0250	0.0250	0.0230
50	MI + closed form	1.59	1.59	1.59	1.61	1.60	1.60
50	Reweighting	0.289	0.290	0.290	0.320	0.320	0.321
100	Classical ML	0.0330	0.0320	0.0330	0.0340	0.0340	0.0340
100	MI + closed form	2.17	2.17	2.18	2.22	2.21	2.21
100	Reweighting	0.610	0.611	0.611	0.677	0.677	0.678
1000	Classical ML	0.286	0.283	0.285	0.307	0.307	0.307
1000	MI + closed form	17.1	17.1	17.1	18.1	18.0	18.0
1000	Reweighting	15.9	15.9	15.9	17.8	17.7	17.7

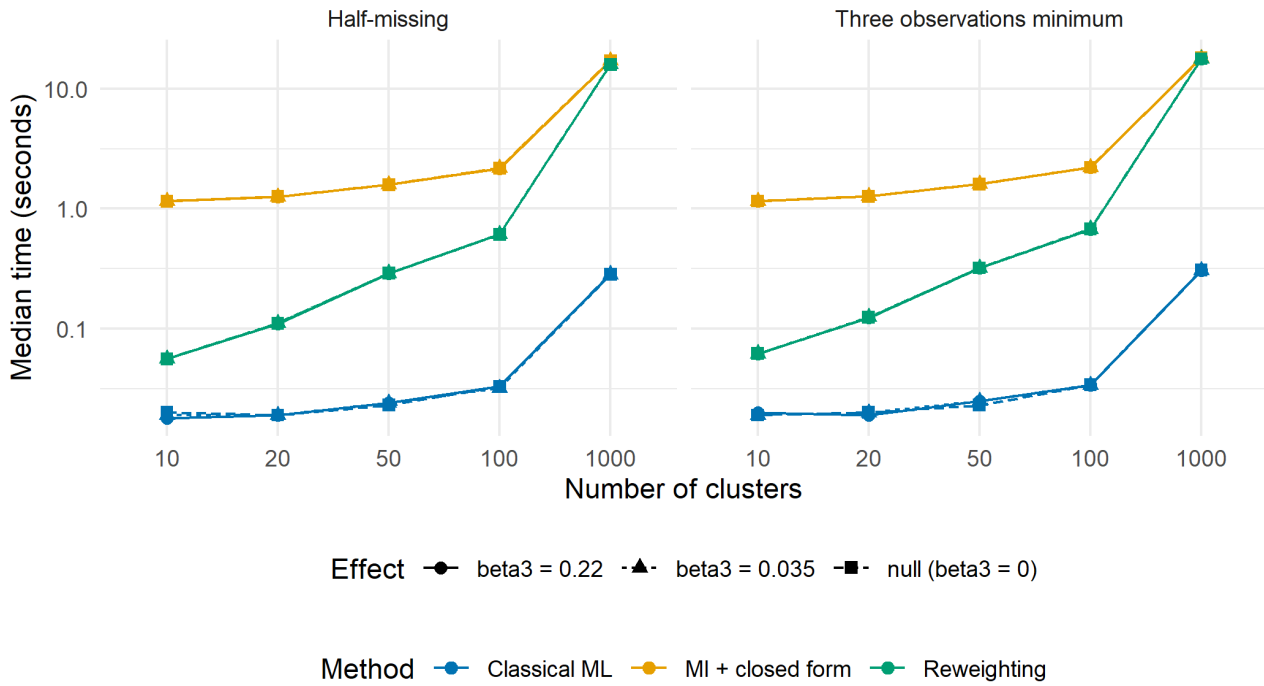


Figure 7: Median computation time per replicate for the linear time trend (log scale).

- Classical ML is fastest by far at every n .
- MI is slowest up to $n = 100$; at $n = 1000$ MI and reweighting are similar.

Fraction of missing information (MI)

Table 11: Mean fraction of missing information for beta3 (MI only).

n	HM: 0.22	HM: 0.035	HM: 0	3O: 0.22	3O: 0.035	3O: 0
10	0.231	0.223	0.226	0.212	0.211	0.205
20	0.252	0.245	0.246	0.230	0.231	0.223
50	0.264	0.262	0.268	0.242	0.239	0.239
100	0.275	0.271	0.271	0.248	0.244	0.243
1000	0.276	0.276	0.277	0.253	0.253	0.251

- The fraction of missing information is higher under half-missing than under three observations minimum, and increases slightly with n .

Results - log time trend

- MSE and coverage are not reported: the linear methods estimate a linear slope, which has no true value under the log trend.
- Effect sizes: large ($\beta_1 = -2.42$, $\beta_3 = 1.95$), small ($\beta_1 = -0.385$, $\beta_3 = 0.31$).

Type I error

Table 12: Type I error rate of the interaction test (identical arms).

n	Method	HM: 0	3O: 0
10	Classical ML	0.153	0.141
10	MI + closed form	0.104	0.0902
10	Reweighting	0.0905	0.0865
20	Classical ML	0.0904	0.0978
20	MI + closed form	0.0744	0.0756
20	Reweighting	0.0614	0.0668
50	Classical ML	0.0646	0.0630
50	MI + closed form	0.0608	0.0606
50	Reweighting	0.0444	0.0438
100	Classical ML	0.0528	0.0532
100	MI + closed form	0.0532	0.0522
100	Reweighting	0.0394	0.0390
1000	Classical ML	0.0524	0.0510
1000	MI + closed form	0.0542	0.0480
1000	Reweighting	0.0368	0.0358

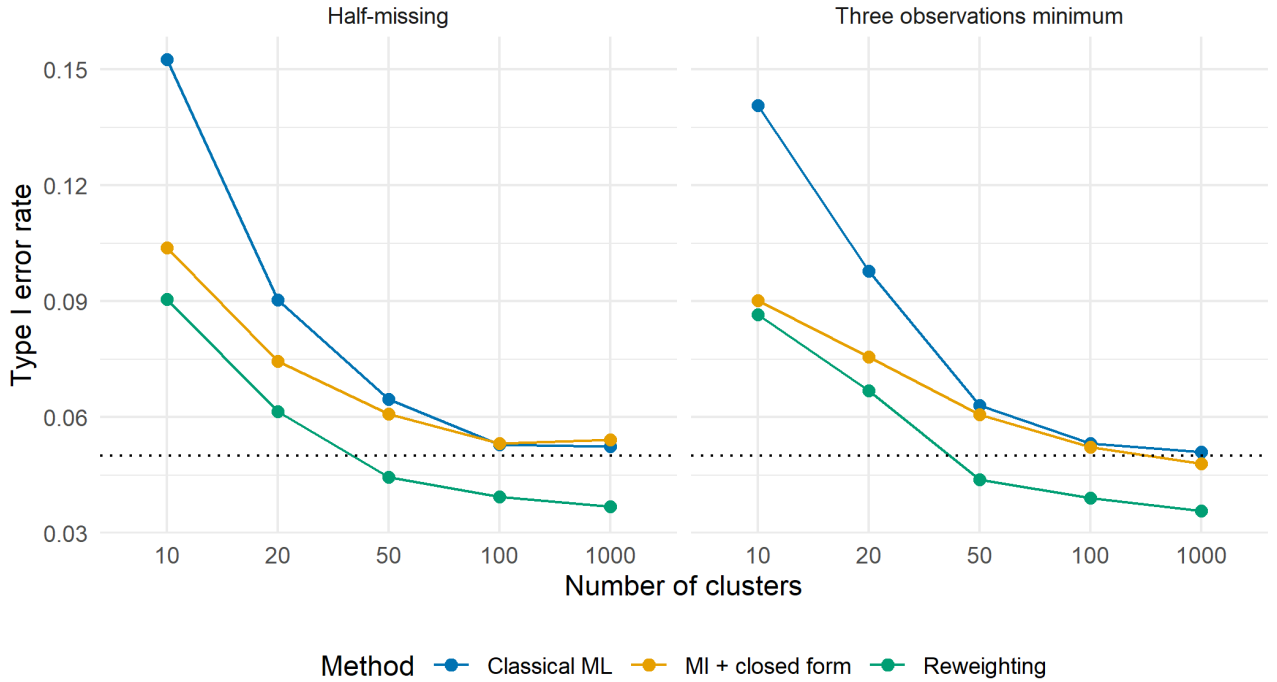


Figure 8: Type I error rate for the log time trend (dotted line: nominal 0.05).

- Small- n inflation as in the linear scenarios; classical ML is again the most inflated.

- Reweighting becomes conservative from $n = 50$ onwards and stays below nominal at $n = 1000$; classical ML and MI approach nominal.

Power

Table 13: Power of the interaction test.

n	Method	HM: large	HM: small	3O: large	3O: small
10	Classical ML	0.363	0.162	0.372	0.146
10	MI + closed form	0.274	0.108	0.279	0.0954
10	Reweighting	0.252	0.0928	0.268	0.0876
20	Classical ML	0.474	0.109	0.501	0.0964
20	MI + closed form	0.408	0.0846	0.423	0.0744
20	Reweighting	0.381	0.0764	0.418	0.0692
50	Classical ML	0.796	0.0910	0.837	0.0870
50	MI + closed form	0.742	0.0912	0.781	0.0758
50	Reweighting	0.737	0.0650	0.781	0.0620
100	Classical ML	0.976	0.108	0.982	0.108
100	MI + closed form	0.955	0.0966	0.964	0.101
100	Reweighting	0.960	0.0830	0.974	0.0828
1000	Classical ML	1.00	0.520	1.00	0.544
1000	MI + closed form	1.00	0.468	1.00	0.482
1000	Reweighting	1.00	0.459	1.00	0.488

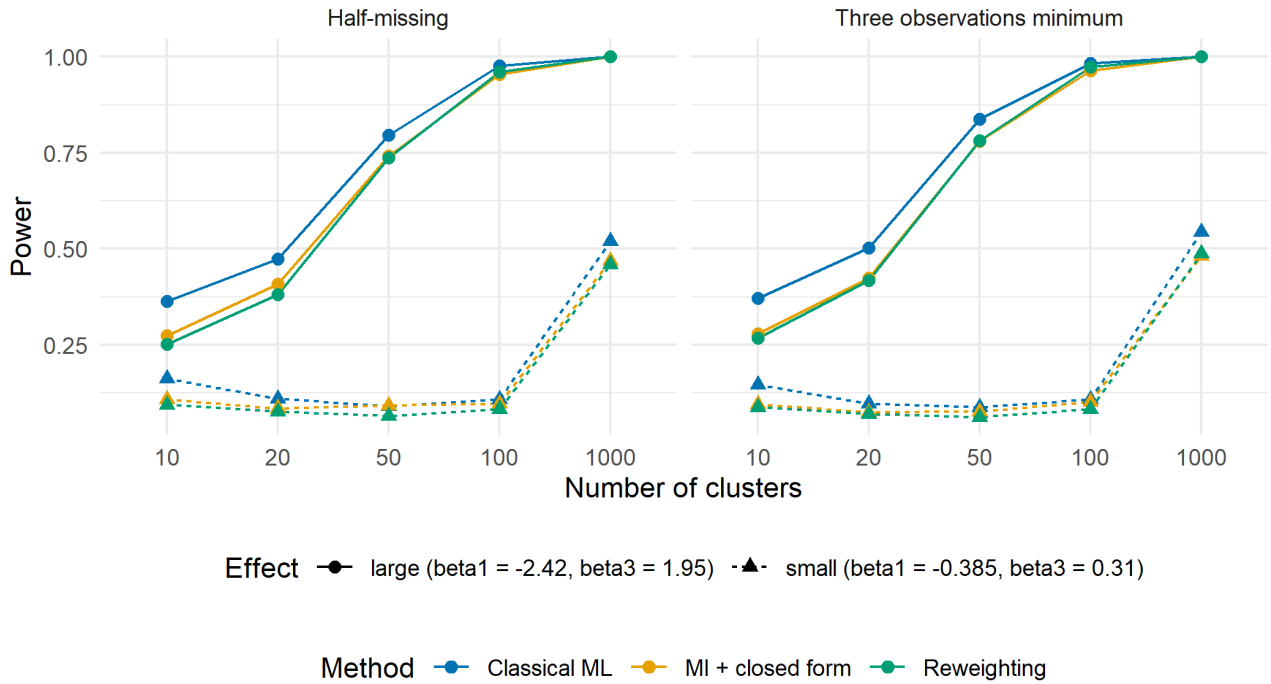


Figure 9: Power for the log time trend.

- The linear methods still detect the crossing well: power is higher than in the linear scenarios with the same end-of-study difference (large vs. $\beta_3 = 0.22$, small vs. $\beta_3 = 0.035$).
- Classical ML has the highest power; MI and reweighting are similar.

Convergence



Figure 10: Convergence status for the log time trend, pooled over effect sizes.

- Same pattern as the linear scenarios, with more singular fits at $n = 10$.
- Classical ML's "not converged" share at $n = 1000$ is again the lme4 gradient false positive.

Computation time

Table 14: Median computation time per replicate (seconds).

n	Method	HM: large	HM: small	HM: 0	3O: large	3O: small	3O: 0
10	Classical ML	0.0180	0.0180	0.0180	0.0200	0.0190	0.0180
10	MI + closed form	1.13	1.14	1.13	1.14	1.14	1.14
10	Reweighting	0.0560	0.0550	0.0550	0.0620	0.0620	0.0620
20	Classical ML	0.0200	0.0200	0.0200	0.0190	0.0180	0.0200
20	MI + closed form	1.23	1.23	1.23	1.24	1.24	1.24
20	Reweighting	0.110	0.110	0.109	0.123	0.123	0.123
50	Classical ML	0.0240	0.0240	0.0240	0.0240	0.0240	0.0230
50	MI + closed form	1.55	1.54	1.54	1.56	1.56	1.56
50	Reweighting	0.287	0.287	0.288	0.320	0.319	0.320
100	Classical ML	0.0310	0.0310	0.0310	0.0320	0.0320	0.0320
100	MI + closed form	2.11	2.11	2.10	2.14	2.14	2.14
100	Reweighting	0.610	0.607	0.608	0.676	0.675	0.675
1000	Classical ML	0.257	0.263	0.265	0.280	0.285	0.287
1000	MI + closed form	16.5	16.4	16.3	17.2	17.1	17.1
1000	Reweighting	15.9	15.8	15.8	17.7	17.6	17.6

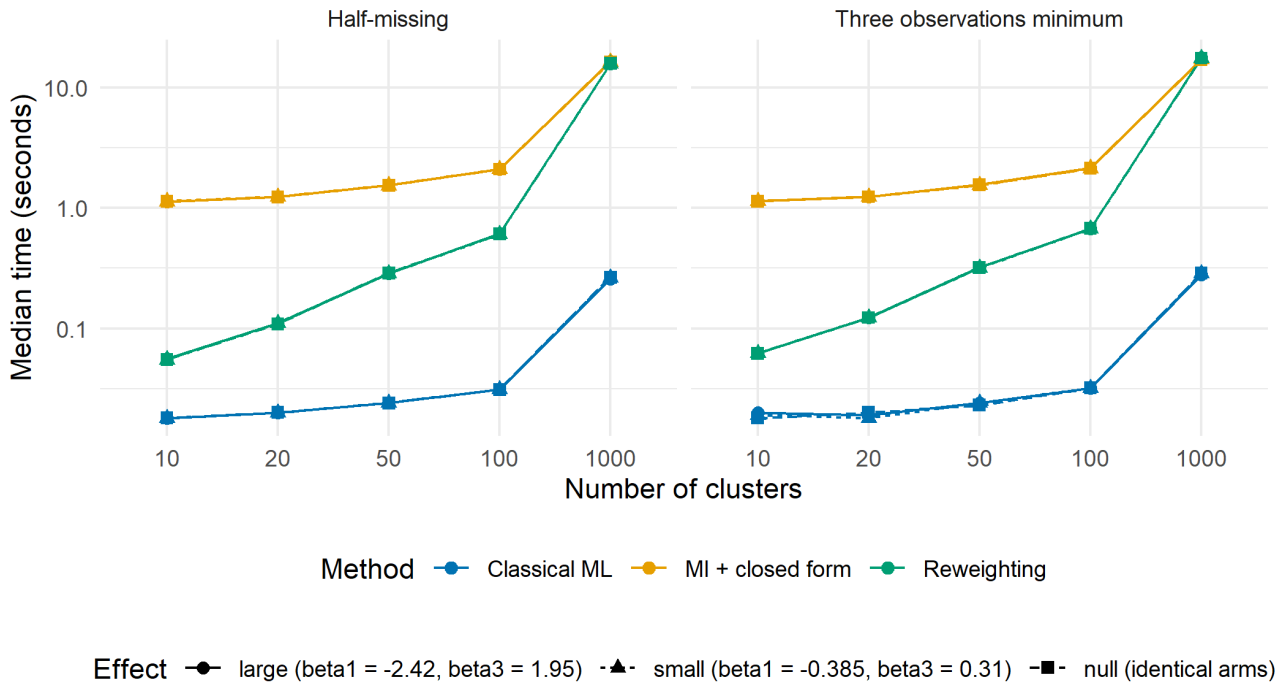


Figure 11: Median computation time per replicate for the log time trend (log scale).

- Same ordering as the linear scenarios.

Fraction of missing information (MI)

Table 15: Mean fraction of missing information for beta3 (MI only).

n	HM: large	HM: small	HM: 0	3O: large	3O: small	3O: 0
10	0.245	0.237	0.238	0.224	0.218	0.220
20	0.270	0.269	0.263	0.246	0.240	0.238
50	0.290	0.281	0.278	0.257	0.255	0.254
100	0.288	0.294	0.291	0.261	0.263	0.263
1000	0.296	0.298	0.296	0.267	0.270	0.271

- Slightly higher than in the linear scenarios; same pattern over dropout mechanism and n .

Suggested improvements

1. LSPIM: make it scale beyond 50 clusters.
2. Small-sample degrees of freedom: Satterthwaite for the classical ML test, Barnard-Rubin for the Rubin-pooled MI test; this targets the type I error inflation at small n .
3. Classical ML convergence: replace lme4's absolute gradient flag with a relative gradient check, to remove the false “not converged” at large n .